NUS GRIP
Ambient Systems

Assignment #2

September 8, 2020

Assignment 1 Revisited



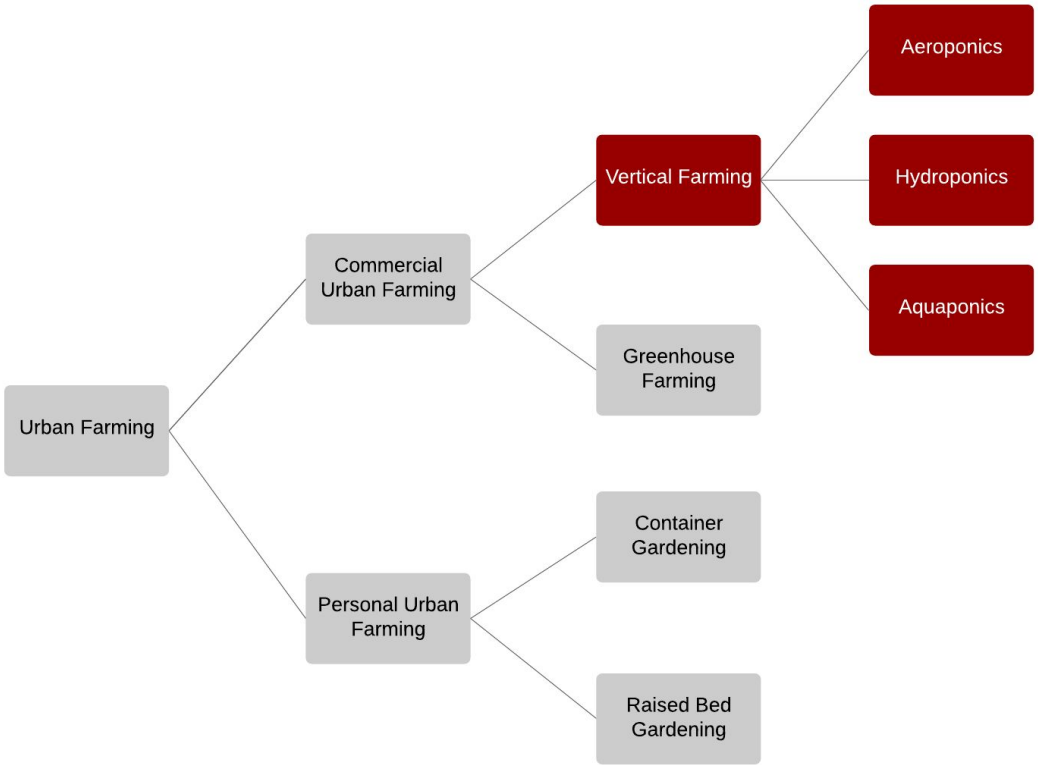
Introduction

Vision: create a safe, healthy, and productive indoor environment for all.

Mission: become a leading global provider of building management and automation systems.

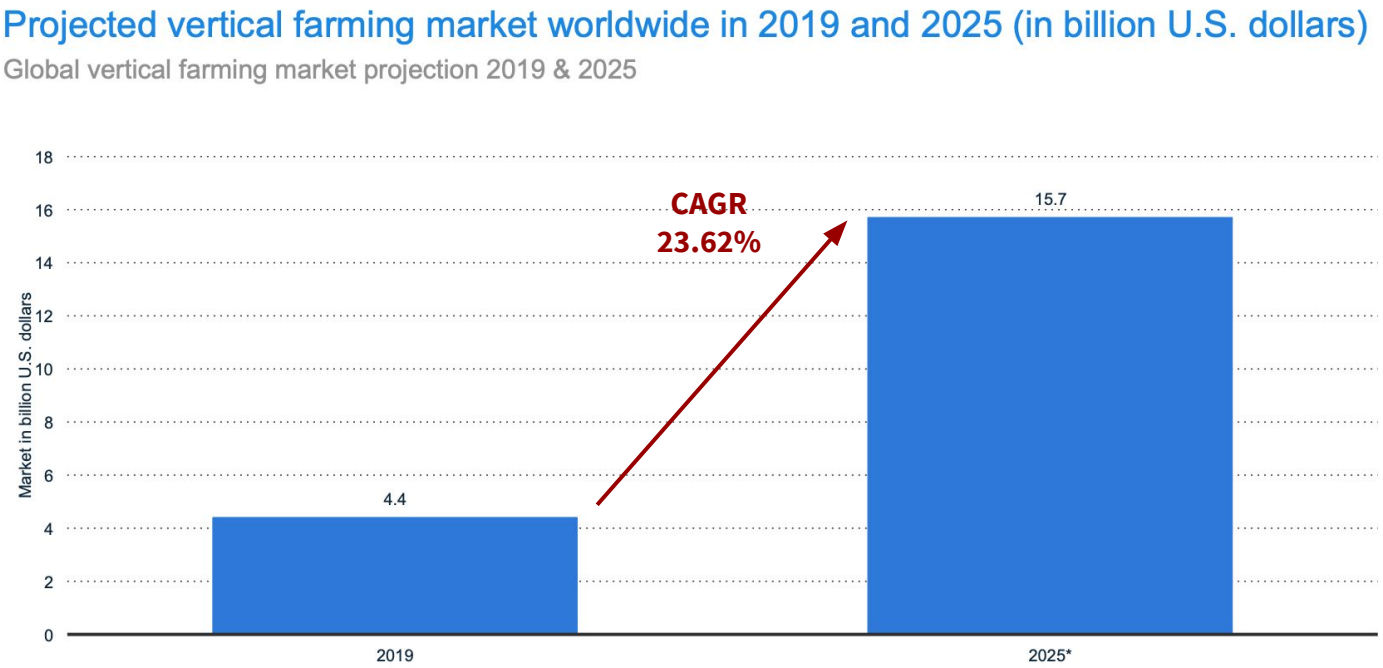
Goal: develop intelligent building automation solutions enabled by IoT and that delight property owners and building operators, starting with the indoor farming segment.

Market Segmentation



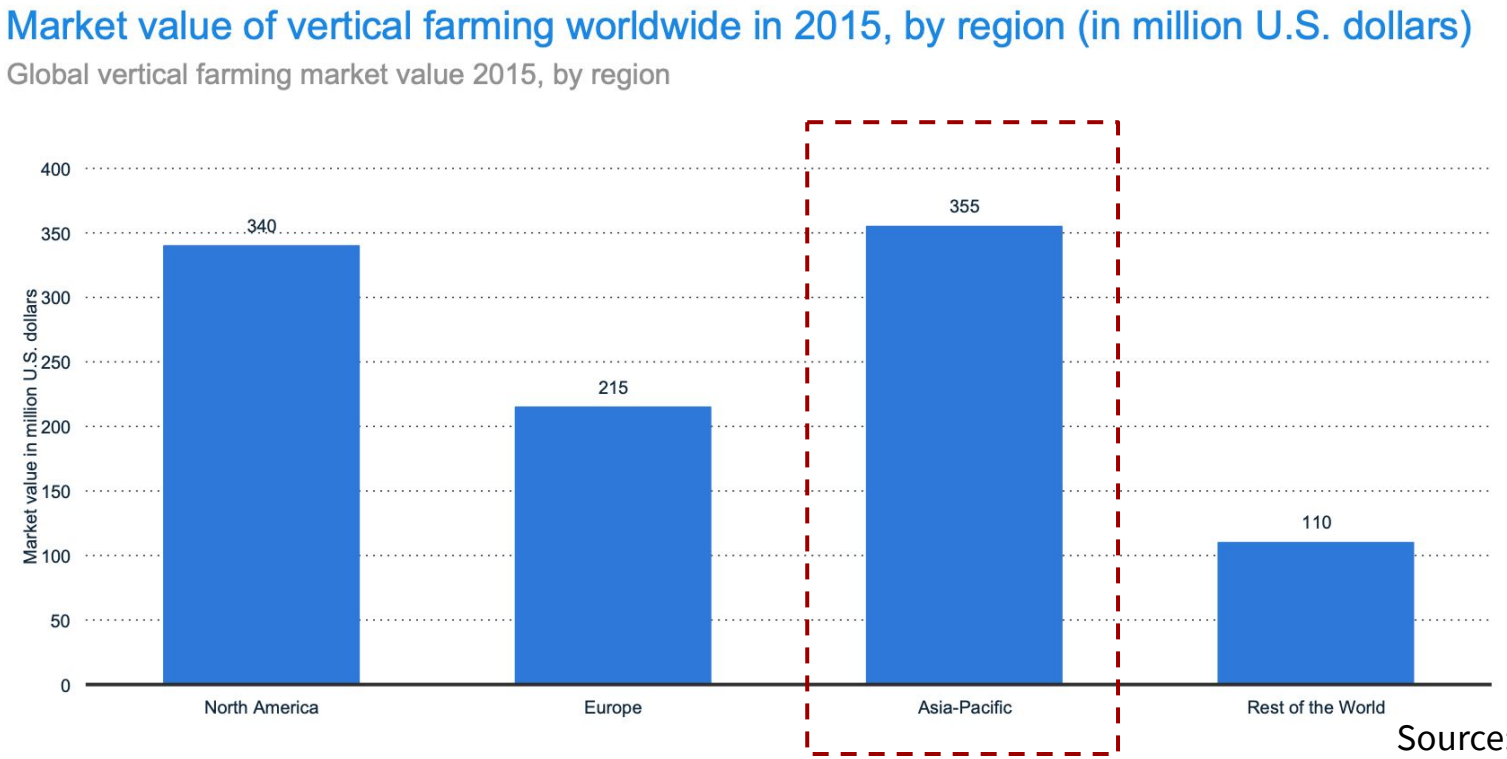
Source: Frost & Sullivan

Vertical Farming Market



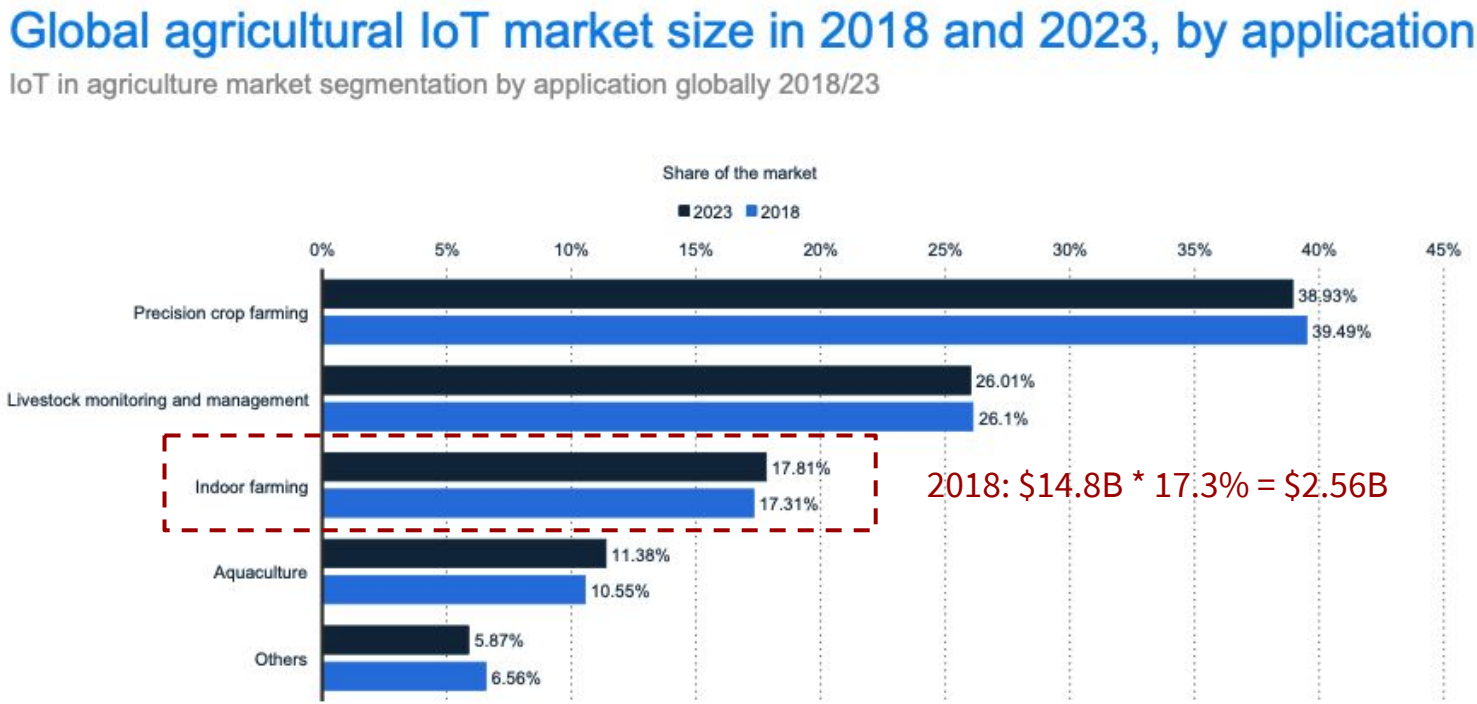
Source: Statista

Vertical Farming Market



Source: Statista

Agricultural IoT Market



2018: $14.8B * 17.3% = $2.56B

Source: Statista

Vertical Farming is a new and rapidly growing industry.

The potential market that can be addressed is greater than $1B.

We will rely on primary market research for further validation.

Section 1: MVP



MVP: “Climate-as-a-Service”

Components:

1. Wireless that measures and manages four main parameters: temperature, moisture, light, and CO2
2. Connection with existing systems
3. User interface delivered via web and mobile platform

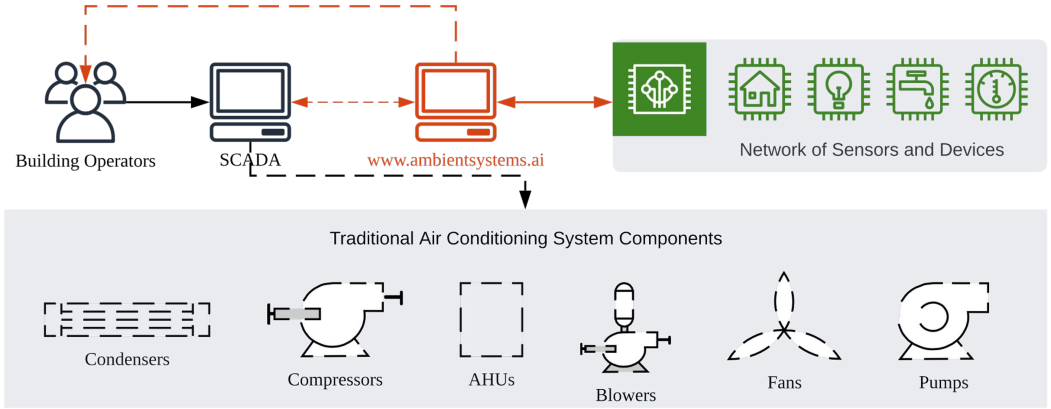
Advantages:

1. Opens up the market (major expense for growers)
2. Minimizes chance of replications
3. Simple streamlined process

Ambient Systems MVP High-level Schematic

What makes Ambient different?

- Technology tailored for vertical farming requirements
- Seamlessly connects to the building's existing systems, including SCADA, and provides notifications directly to the operators
- Managed on the cloud and available across multiple devices
- Enabled by a network of interconnected sensors and devices (Internet-of-Things)
- Leverages NUS intellectual property to minimize maintenance and energy costs of sensors and devices



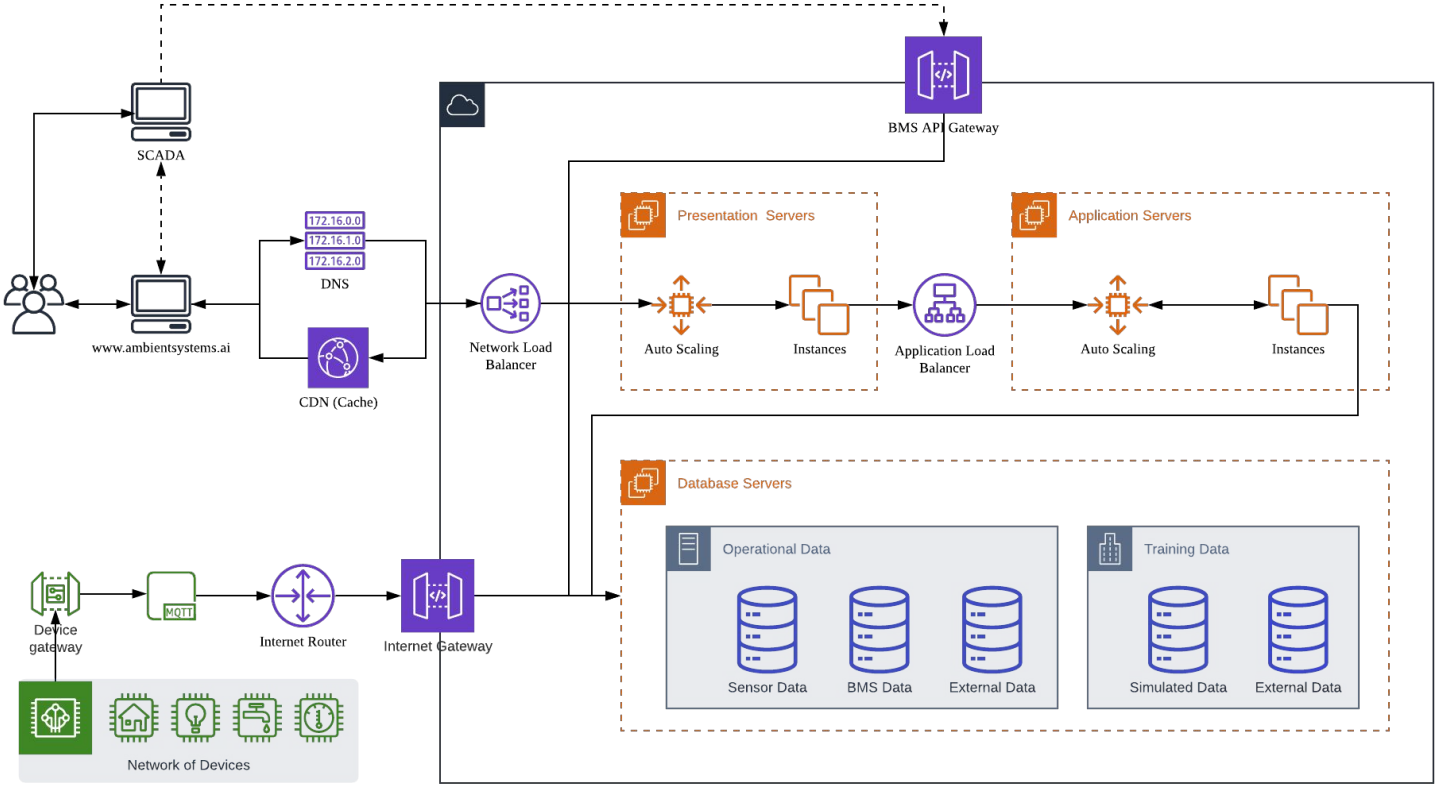
Section 2: Product Specifications



Components

Proprietary Hardware (IoT Network)	Commercially Available Technology	Proprietary Software (Platform and Firmware)
• Integrated circuit design • Power source • Micro-processor (controller) • Transceiver • Memory • Temperature sensor • Moisture (humidity) sensor • Light sensor • Chemical sensors (e.g., CO2) • Network gateway	• Power source • Cables • Device mounting mechanisms • (Optional) HVAC system • (Optional) LED lighting • (Optional) Electrical • (Optional) Security and CCTV	• User interface delivered via web and mobile dashboard • Secure and private IoT device management platform • Secure and private IoT system communication protocols • Scheduling mechanism • Calibration and optimization models • Notification and alert system • API design and inter-connection with existing systems (e.g., farm management, lighting, security)

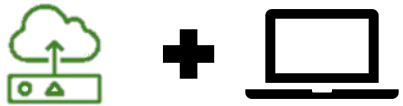
Preliminary System Architecture



Section 3: Product Roadmap



Product Roadmap

and Platform Aug 2021	ML-enabled Optimization Jan 2022	Controls Aug 2022
First product milestone, which is also the MVP, will consists of a wireless network of sensors and a platform to display this data in one place.	Second product milestone will include a layer of added intelligence to the existing system in order to provide recommendations for enhancing operations.	Third product milestone will include a layer of controls that will enable fine-tuning of climate parameters remotely and automatically to create optimal conditions.
 Sensors + Visibility	 Sensors + Visibility Recommendations	 Sensors + Visibility + Actuators Recommendations Automation

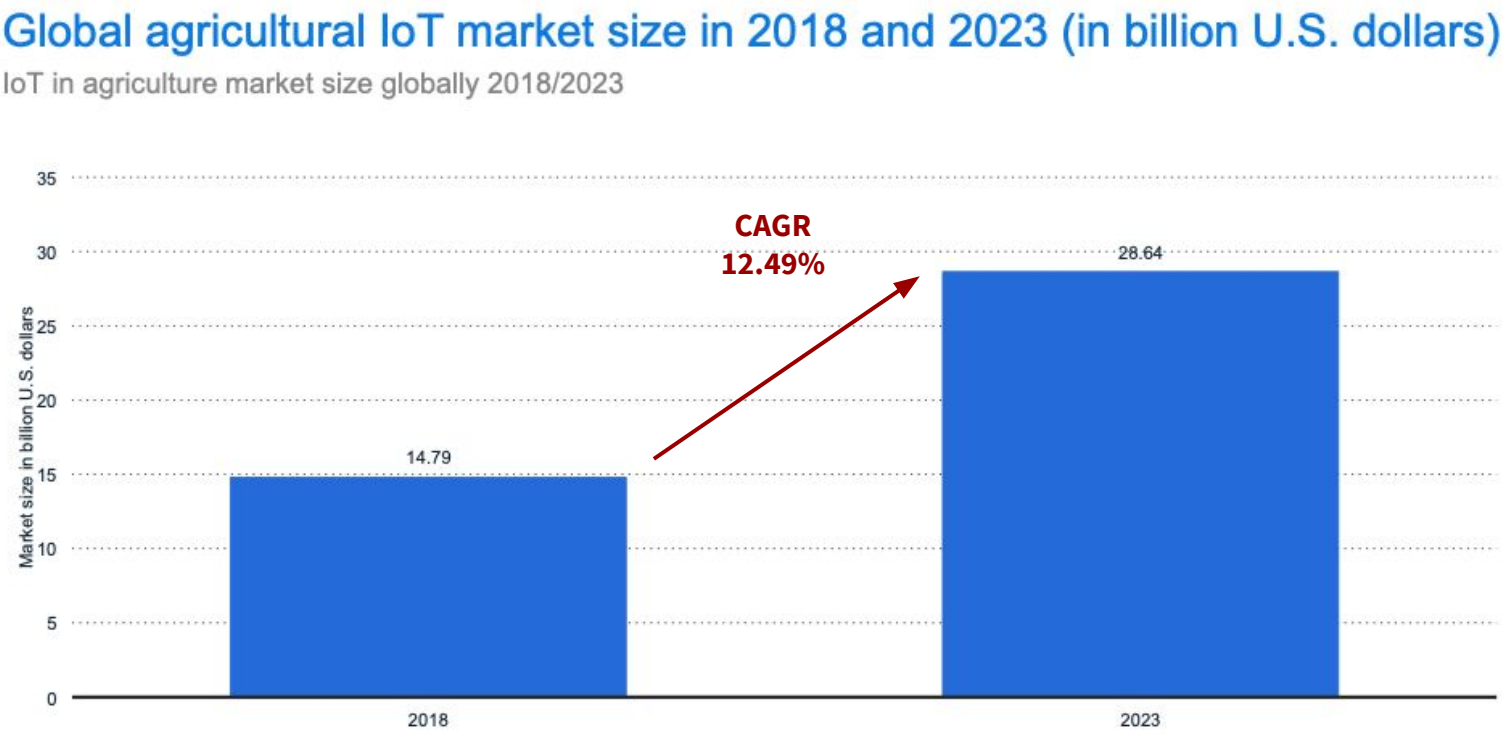
Thank you

Appendix

Sample Companies

- www.plenty.ag
- www.cropx.com
- www.onfarm.com
- www.farmobile.com
- www.artemisag.com
- www.upgrownfarming.co
- www.skygreens.com
- www.susteniragriculture.com
- www.life3.co
- www.vertivegies.farm
- www.pfsap.panasonic.com.sg/ps_client_agriculture.aspx

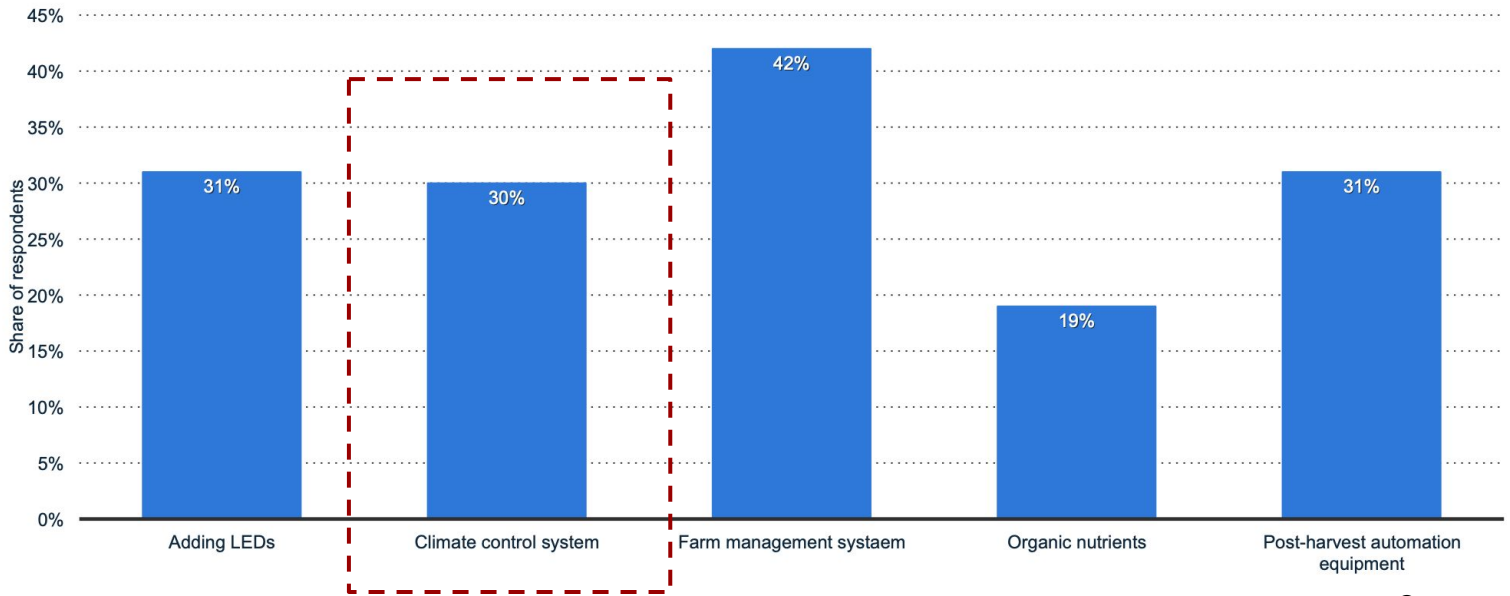
Agricultural IoT Market



Vertical Farming Market

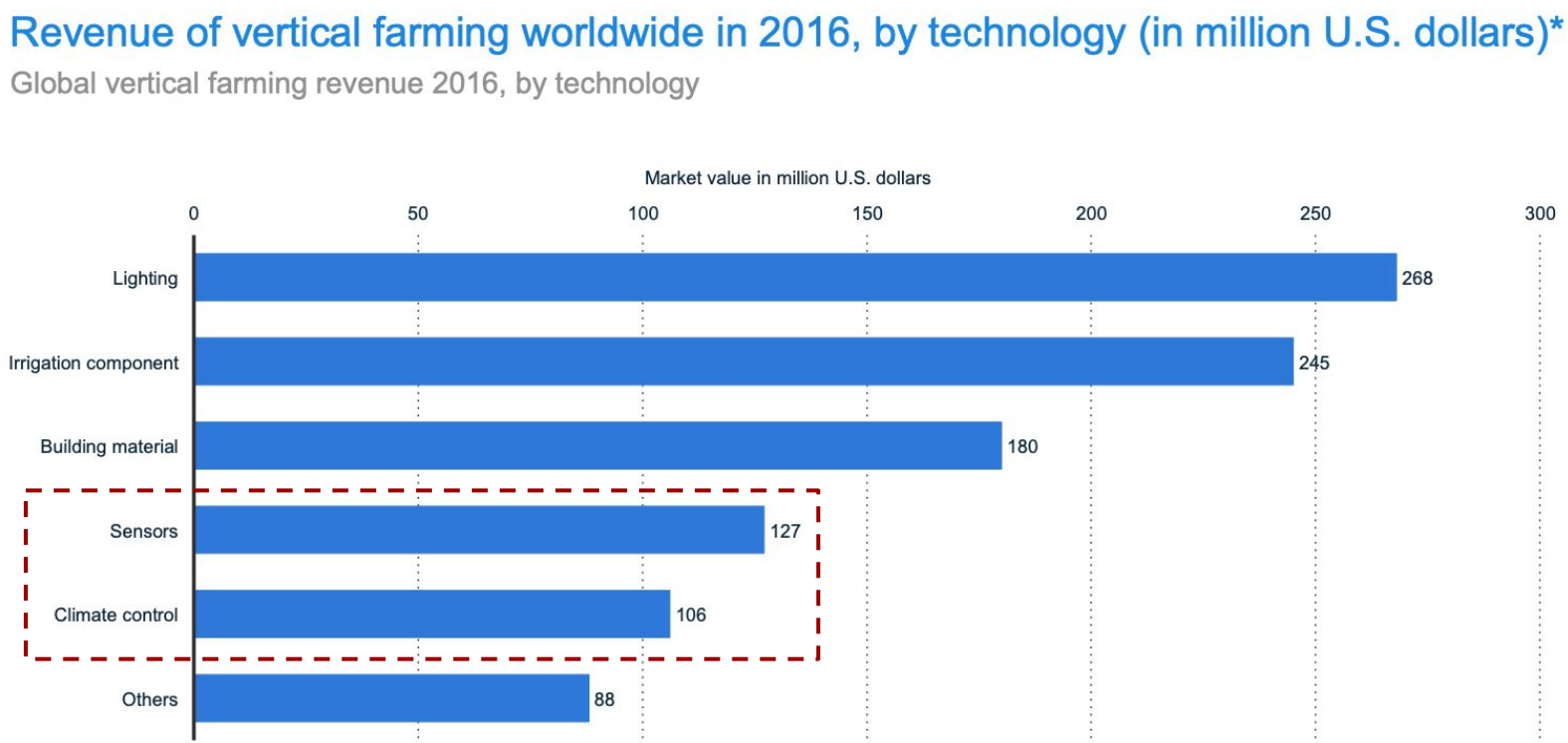
Intended technology and farm improvements in indoor farming in the next year worldwide as of 2016

Planned improvements in indoor farming worldwide 2016



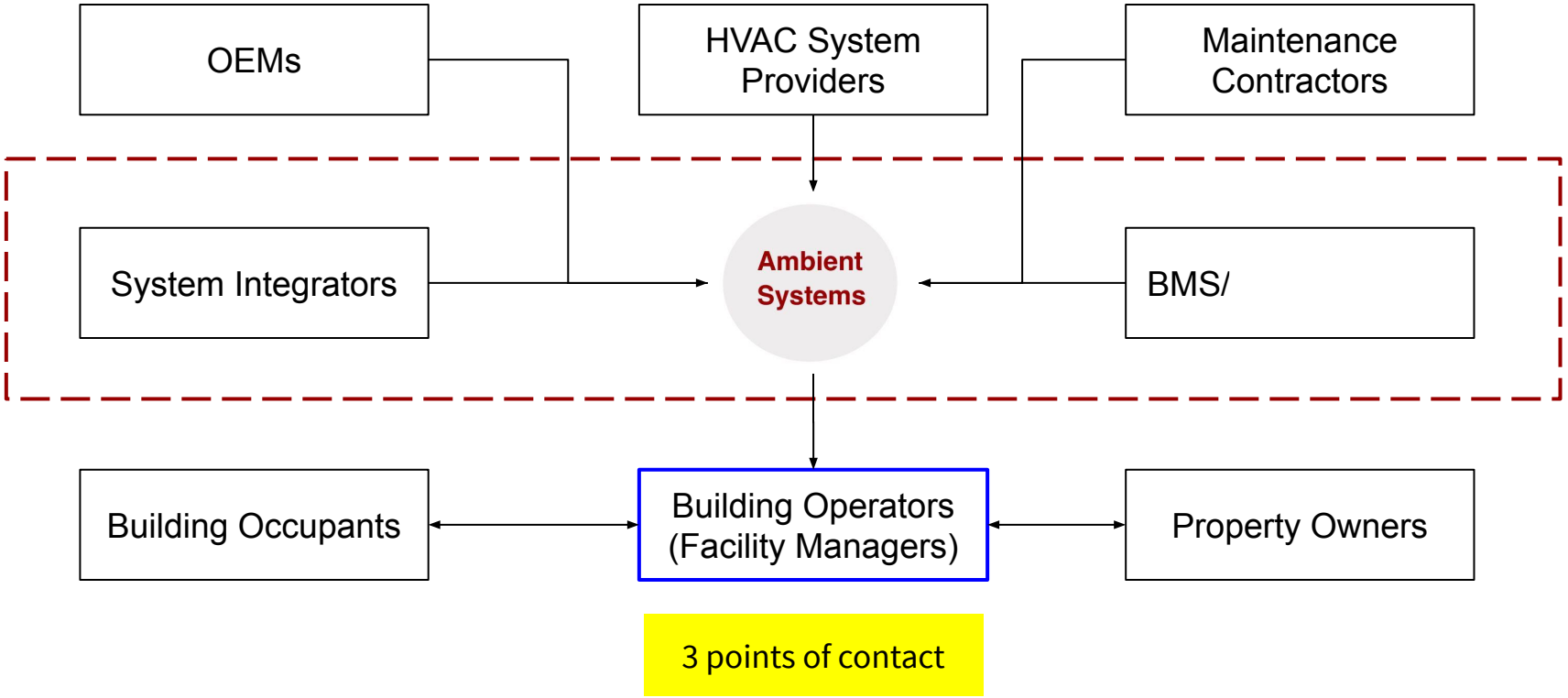
Source: Statista

Vertical Farming Market

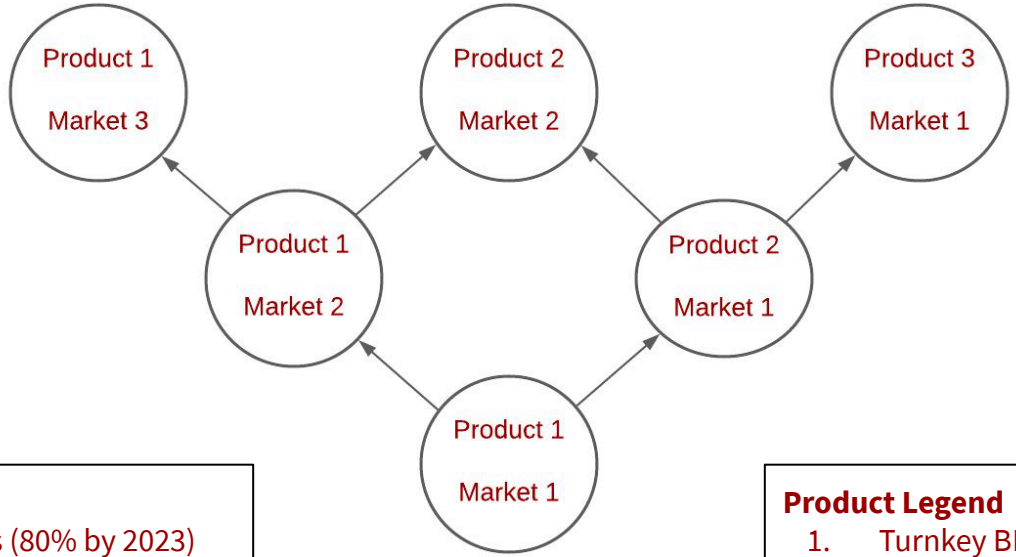


Source: Statista

Value Chain Analysis



Bowling Pin Diagram



Market Legend

1. M1: SG Indoor Farms (80% by 2023)
2. M2: APAC C&I (50% by 2029)
3. M3: Europe C&I (50% by 2031)

Product Legend

1. Turnkey BMS (system integrator)
2. enabled BAS
3. enabled

3s Framework

